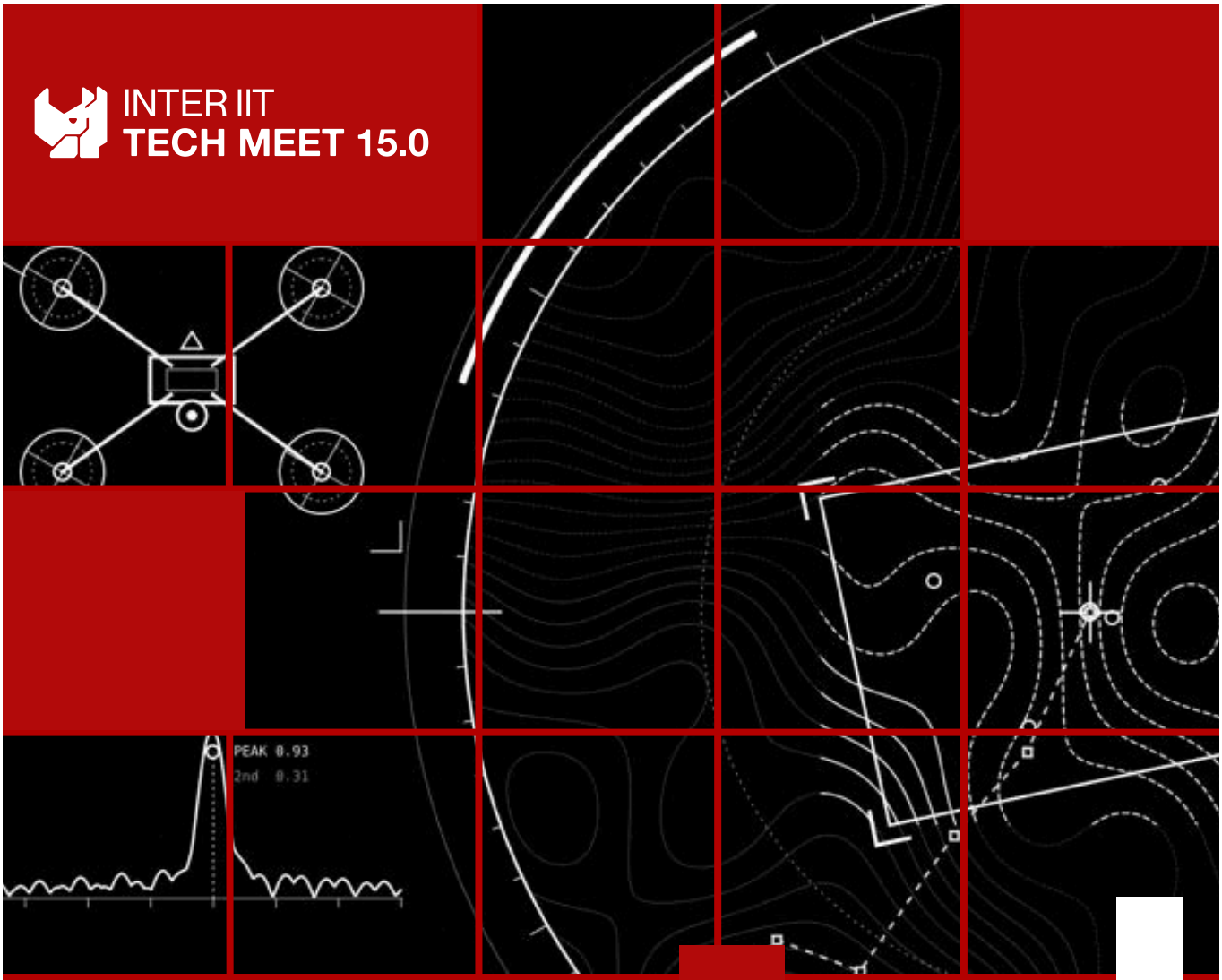
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TECH MEET 15.0




# HIGH PREP

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Create. Inspire

# UAV Visual Geo-Localisation Challenge

# About ideaForge

At the forefront of cutting-edge drone technology, with a focus on developing superior solutions for intelligence, surveillance, and reconnaissance (ISR) purposes, ideaForge has its origins nearly two decades ago. The company is built with a mission to develop technically superior UAVs for various missions, utilizing modern advancements in aerospace and robotics. From being a humble start-up, now ideaForge Technology has recently offered its IPO and become a listed company in NSE and BSE.

ideaForge is a vertically integrated company with an in-house product development centre, allowing us to design, develop, engineer, and manufacture indigenous unmanned aerial vehicles (UAVs). ideaForge was founded in 2007, and over time became the pioneer and pre-eminent market leader in the Indian unmanned aircraft systems ("UAS") market. ideaForge had the largest operational deployment of indigenous UAVs across India, with an ideaForge manufactured drone taking off every four minutes on average for surveillance and mapping. As of today, their customers have completed over 500,000 flights using our UAVs.

ideaForge UAVs are equipped with industry-leading specifications and capabilities, comparable to those of other established global players in the UAV industry. Their in-house design, development, engineering, and manufacturing capabilities have enabled us to develop better products in response to changing customer demands, thereby improving customer experience with our products. The company serves domestic and international customers in the defense and civil sectors, primarily for surveillance, mapping, and surveying applications. Our customers include the Indian Armed Forces, homeland security authorities comprising certain central armed police forces, state police departments, disaster management forces, forest departments, government along with private entities, enterprise sectors and many more.

The company is closely working with the Indian Armed Forces for the deployment of their drones for important security and surveillance missions. ideaForge UAVs are deployed across Indian borders, assisting the Indian Army in surveillance and secure the nation.

Furthermore, several Police Departments, Government and private authorities have deployed the ideaForge UAVs for a variety of surveillance, mapping, and surveying operations such as anti-terror, anti-Naxal operations, illegal activity prevention, crime control, traffic monitoring, mine area planning, volumetric estimations, land survey and many others.

# Background and objective

UAVs for outdoor wide area surveillance depend on GNSS for position and navigation. When GNSS is jammed, spoofed or unavailable, matching the onboard camera view to stored satellite imagery is one of the few ways to recover an absolute position fix. The hard cases are thermal imagery matched against daytime satellite maps, seasonal change, low-texture terrain, and a tight compute budget to run on the UAV using edge processing.

In this challenge, teams will build a system that locates a UAV image on a satellite map, for both daytime and thermal imagery, running entirely on an embedded computer.

The goal is a robust, self-aware localizer: accurate where a match is possible, and willing to decline where it is not. A confident wrong answer is treated as worse than no answer.

# Problem Statement

The problem involves a set of UAV images and an approximate location for each. The final evaluation dataset will use a reference dataset region of around 100km x 100km and the approximate location (search radius) will be within 5km of the actual ground truth. Additional metadata related to the captured images such as UAV attitude and altitude as well as details about camera parameters such as resolution, FOV, intrinsics, image formats etc will also be provided. The objective is to calculate the geo-coordinate of the center position of each image. Each image may not resolve into a high confidence match, hence certain images could also return a “no match” status. The scoring will assign higher scores to more accurate geo-locations and in cases of very high error - negative scores will also be assigned.

**Evaluation Data Mix.**

The evaluation set will be 50% EO and 50% IR images. Since most of the commonly available satellite imagery datasets use EO cameras, the matching of IR images against EO reference data would be more challenging. Input images will also be spread across multiple altitudes above ground and may go all the way up to 3000m AGL.

# System assumptions

Model training, data pre-processing and development testing may happen on any system available to the teams. However, in the final evaluation all solutions must run on the fixed target hardware which would be an NVIDIA Jetson Orin NX 16 GB with 512GB NVMe SSD.

# Competition phases and dataset

The competition runs over 8 weeks in three phases, ending with a blind evaluation on a region teams have not seen.

**1. Problem Definition and Prototype Development (weeks 1–2)**
The complete details of the problem which will complement this high level problem specification - including the exact input and output formats, altitude limits, satellite datasets to be used, hardware and software configurations of the embedded compute module to be used in evaluation, exact scoring metrics, etc will be communicated by the end of Week 1. The reference datasets for training and internal evaluation will be provided by the end of Week 2.

**2. Mid-point review (end of week 4)**
Teams should submit a brief design note on their solution approach as well as a working container that runs on the target hardware and software configuration. This will ensure that solutions and the input output interfaces fit the platform early on and avoid rework at the final evaluation stage.

**3. Final evaluation (end of week 8)**
For the final evaluation, the reference matching region will be provided to the teams 24-48 hours before the evaluation deadline. Teams can use this advance preparation period to pre-process data or generate any artefacts which need to be loaded on the compute platform for matching during the evaluation. The blind evaluation dataset will be provided on the day of the final evaluation. Depending on the logistics feasibility of the final evaluation phase, teams may be asked to submit the container of their evaluation to be run on the organizer's hardware or asked to run the container on their own hardware conforming to the same specs, in the presence of the organizers and generate the results artefacts.

# Rules of engagement

Solutions must be fully automatic, self-contained and reproducible on the target hardware.

**Submission**
• One Docker image built on the target hardware and specified base image, with all models and code/scripts inside it.
• A single entry point that takes the reference data folder, the input folder and an output path.
• Output in the specified format: one row per image with name, status, latitude, longitude, (if matched) and an optional confidence score.
• A technical report describing full details of the method, data used, any internal benchmarking results on public or test datasets and known limitations.

**Allowed**
• Any method or model, including pretrained models with a license that permits this use declaring the public pre-trained models used in the final report.
• Separate models or methods for EO and IR imagery as long as the complete solution fits within the hardware limits of the embedded platform. • Teams may preprocess the final evaluation region's reference data on any hardware with up to 1 GPU, within the advance window. Preprocessing must be fully automatic, using code, weights and configuration submitted before the region is released. No manual annotation, tuning or code changes are permitted after release. Organisers may re-run the submitted pipeline to verify the results.

**Not allowed**
• Network access during evaluation, or downloading any data at run time.
• Attempts to reverse engineer ground truth from file names, ordering, timestamps, etc.
• Any attempt to game the evaluation method by skipping entire segments of evaluation data.
• Using any reference or training imagery of the evaluation region other than what organisers supply.

**Verification.**
Organisers may inspect the code, execution logs of any top-ranked submission and re-run it. A violation leads to disqualification.

**Data handling.**
All supplied imagery is for this competition only, may not be redistributed, and must be deleted afterwards.

# Evaluation Dataset Scoring Methodology

The experimental evaluation score on the blind dataset will consider multiple factors to arrive at a consolidated score:

**Position Error:** Error is the ground distance in metres between the predicted and true position. Depending on the camera type and altitude bands, error ranges will be declared - a prediction in the lowest error range will earn the highest positive points, next higher error range will earn lower positive points. No match will earn 0 points and an error greater than a specified threshold will earn negative points. For example, a representative scoring might be

| Prediction Accuracy | Points |
|---|---|
| Error under 100 m | +8 |
| 100–250 m | +4 |
| 250–500 m | +2 |
| Over 500 m | -4 |
| No match returned | 0 |

**Minimum Availability:**
In a practical UAV localization scenario, a good solution must be available across all operating conditions. Hence, to avoid teams skipping some parts of the more challenging evaluation zones such as IR imagery or higher altitudes by always predicting “no match” to avoid any negative scores, we may enforce a minimum prediction rate or availability for each group.

**Latency:**
Latency will be measured as the total time taken to process the entire final blind evaluation dataset on the target hardware. Total latency, combined with availability will prevent teams from skipping or returning “no match” on harder datasets.

**Final Evaluation Dataset Score:**
The final score will be calculated by weighting the total error, availability and latency parameters of the final evaluation.

# Final Contest Scores

The final score for the entire problem duration will be composed of the Mid-term and Final presentations and the final evaluation dataset score

| Component | Weight |
|---|---|
| Mid-point review | 20% (based on clarity of solution approach, sample container complying to input output interfaces |
| 100–250 m | 20% (based on explanation of solution approach, presentation and response to jury questions) |
| 250–500 m | 60% (based on the final weighted score considering the accuracy, availability and latency criteria published) |

*Note: The exact and full problem specifications including the image resolutions, FOVs, input formats, altitude ranges, the hardware and software platform specifications and the scoring metrics including positive and negative weightages for different input conditions, criteria for minimum availability and latency calculations will be declared by the end of Week 1.*

***Disclaimer:***
*Since this is a high-prep, research oriented problem, not all aspects of the constraints, assumptions, criteria etc. may be fully detailed in the limited problem statement text - certain specifics may be missing from the problem specification which may be clarified over the Discord group communications or during any review. In case of any such findings, the organizers reserve the right to add these clarifications to the problem statement with the stipulation that all changes or clarifications will be communicated at the same time to all teams even if the query is raised by one team and that these clarifications do not materially alter the nature of the problem itself. The final decision in this regard will rest with the organizers themselves and cannot be challenged by any of the teams.*